EE-170L Computer Programming Lab

Laboratory Report — No. 05

The while Loop, Nested while Loop, and do...while Loop in C++

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1. Lab Objectives

By the end of this lab session, students will be able to:

- Understand and apply the while loop for condition-based repetition.
- Use nested while loops to produce multi-dimensional output patterns.
- Understand and apply the do...while loop and explain its key difference from while.
- Choose the appropriate loop type for a given programming problem.

2. The while Loop

A while loop is a control flow statement that allows code to be executed repeatedly based on a given condition. It consists of a block of code and a condition. The condition is first evaluated, and if it is true, the code within the block is executed. This process repeats until the condition becomes false.

2.1 Syntax

```
initialization
while (condition)
{
    statements;
    update variable value;
}
```

Key points about the while loop:

- The condition is checked before every iteration — if false from the start, the body never executes.
- The loop control variable must be initialized before the loop and updated inside the loop body to avoid an infinite loop.

- It is best suited when the number of iterations is not known in advance.

2.2 Example 1 — Print Values from 5 to 1

Write a program using a while loop that prints the values from 5 down to 1:

```
#include <iostream>
using namespace std;
int main()
{
    int i = 5; // loop control variable

    while (i > 0)
    {
        cout << i << "\n";
        i = i - 1;
    }
    cout << "outside loop now";
    return 0;
}
```

2.3 Example 2 — Print Numbers Until -1 is Entered

This program continuously reads numbers from the user and prints them. The while loop exits when the user enters -1:

```
#include <iostream>
using namespace std;
int main()
{
    int num = 5;

    while (num != -1)
    {
        cout << "Enter a number to print: -1 to quit \n";
        cin >> num;
        cout << "You entered: " << num << endl;
    }
    cout << "Outside loop now";
    return 0;
}
```

3. Nested while Loop

A nested while loop is a while loop placed inside the body of another while loop. The inner loop completes all its iterations for every single iteration of the outer loop. This structure is commonly used to print patterns, work with 2D data, or perform repeated grouped operations.

3.1 Example — Descending Number Pattern

Write a program that uses a nested while loop to print the following descending pattern:

```
Output:
    1 2 3 4 5 6 7
    1 2 3 4 5 6
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
#include <iostream>
using namespace std;
int main()
{
    int num = 7;
    int j;

    while (num > 0)
    {
        j = 1;
        while (j <= num) // nested loop
        {
            cout << j << " ";
            j++; // equivalent to j = j + 1
        }
        cout << "\n";
        num--;
    }
    return 0;
}
```

4. The do...while Loop

The do...while loop is a variant of the while loop with one important difference: the body of the loop is executed once before the condition is checked. This guarantees that the loop body runs at least one time, regardless of whether the condition is initially true or false.

Execution flow of the do...while loop:

- The loop body executes first, before any condition check.
- After execution, the condition is evaluated.
- If the condition is true, the loop body executes again.
- The condition is re-evaluated after each iteration.
- The process repeats until the condition becomes false.

4.1 Syntax

```
initialization
do {
    statements;
    update variable value;
} while (condition);
```

Note: The semicolon after while (condition); is mandatory in a do...while loop.

4.2 Example — Password Checker

This program repeatedly prompts the user for a password. The do...while loop ensures the prompt appears at least once, and continues until the correct password (1234) is entered:

```
#include <iostream>
using namespace std;

int main() {
    int password;

    do {
        cout << "Enter password (1234 to exit): ";
        cin >> password;

        if (password != 1234) {
            cout << "Wrong password! Try again.\n";
        }

    } while (password != 1234);

    cout << "Access Granted!" << endl;
    return 0;
}
```

5. Comparison: while vs do...while

Feature	while Loop	do...while Loop
Condition check	Before the loop body	After the loop body
Minimum executions	0 (may never run)	1 (always runs at least once)
Best used when	Iterations may be zero	Loop must run at least once
Semicolon after while	Not required	Required — } while (cond);

6. Conclusion

This lab provided a thorough introduction to loop structures in C++. The three loop types studied — while, nested while, and do...while — each serve distinct purposes in controlling program repetition:

- **while Loop:** Checks the condition before executing the body. Suitable when the number of iterations depends on a condition that may already be false at the start.
- **Nested while Loop:** Places one while loop inside another, enabling complex iterative patterns such as grids and descending sequences.
- **do...while Loop:** Guarantees the loop body executes at least once before the condition is checked. Ideal for input validation and menu-driven programs where a prompt must always be shown.

A solid understanding of these loop structures is essential for solving repetitive computational problems and forms the foundation for more advanced topics such as arrays and functions in future labs.